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By hand:

Step 2: Approach:
In this assignment, the task is to decompress an image that has been given as pixel instructions in an array. The first lines of the code are declaring a grid, and the following lines are ordered, [black, white, black, white], indicating that the first pixel is filled in, in the column, the next is not filled in, so on and so forth. In the code, it will look like [1,1,1,1], one black, one white, one black, one white. The goal here is to make a Python code that can understand the array and print a certain symbol depending on whether the sport is to be "filled in" or not. The first step is to open a JSON file containing the array, then interpret it in a way that will print the image we want.

Step 5: Compare:
Lots of the pros of mine are that it is concise, and it is pretty simple to understand. The pros of the AI-written pseudo code is it is much more detailed, and it better explains that we will be creating a grid with the information we have been given. To improve my pseudocode, I can implement the sections on creating the grid; this way, I can fully map out the area where I will print. To improve on the AI's pseudocode, I would probably just use more of the language style we learned about in the reading, as well as simplify some expressions so it is more concise and easier to read. I believe that the pseudocode is an accurate representation of how this would be programmed into a computer.

Improved pseudocode:

```
OPEN folder
READ contents

grid <- columns * rows

SET current_column <- 0

FOR column in data

    SET current_row <- 0
    filled <- True

    FOR each number in column
        SET count <- 0
        WHILE count != number

            IF filled = True
                PUT "#" into grid[current_row][current_column]
            ELSE
                PUT " " into grid[current_row][current_column]

            SET current_row <- current_row + 1
            SET count <- count + 1

        END WHILE

        SET filled <- NOT filled

    END FOR

    SET current_column <- current_column + 1

END FOR

Print each row
```

```
pseudocode:

OPEN folder
READ contents

FOR column in data
    SET current_row <- 0
    filled <- True

    FOR each number in column
        SET count <- 0
        WHILE count != number

            IF filled = True
                PUT "#"
            ELSE
                PUT " "

            SET current_row <- current_row + 1
            SET count <- count + 1
        END WHILE

        SET filled <- NOT filled
    END FOR

END FOR

Print each row
```

```
AI Pseudocode:

START

Open the JSON file

Read the following information from the file:
    rows
    columns
    data

Create a 2D grid filled with blank spaces
    grid = rows x columns

FOR each column in data

    Set currentRow = 0
    Set filled = TRUE
    # TRUE means place "#"
    # FALSE means place blank space

    FOR each number in the current column

        Repeat number times

            IF filled is TRUE THEN
                Place "#" into grid[currentRow][currentColumn]
            ELSE
                Place blank space into grid[currentRow][currentColumn]
            END IF

            Move to the next row
            currentRow = currentRow + 1

        END REPEAT

        Switch filled to the opposite value
        IF filled is TRUE
            filled = FALSE
        ELSE
            filled = TRUE
        END IF

    END FOR

END FOR

FOR each row in the grid

    Print all characters in that row together

END FOR

END
```